

Barrel shifter

A barrel shifter moves every bit in one operation, logical left, logical right, arithmetic right, or rotate

Stages, modes, result

- Starter: hex D two is one-one-zero-one-zero-zero-one-zero
- Logical left by three enables amount bits one and two
- After stage one you get one-zero-one-zero-zero-one-zero-zero
- Logical right fills with zeros from the left
- Arithmetic right replicates the sign bit
- Rotate wraps bits with no fill loss

Browser lab

Digital Design and Verification Platform

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INTERACTIVE TOOL

Barrel shifter

Shift or rotate an 8-bit word by a select amount — implemented as **log₂N mux stages** (1, 2, 4), not N serial shifters.

Starter example: 0xD2 (11010010) logical left by 3 — stages use amt bits 1+2 (not three serial ×1).

Load starter example

Challenges (1/22)

Quiz: layers. 8-bit barrel needs how many mux stages? Answer: 3

Your answer

Show hintCheckNextIdle

quiz-logquiz-powersquiz-sraquiz-rotstarterstarter-outamt-bits-3stepafter-x1sra-presetrol-presetsrl-zeroamt5explaintogglemode-rorquiz-bypassidentityror-wrapquiz-latencyhex-loadfull-path

Core ideas

Barrel

Any shift 0...N-1 in [log₂N] mux layers.

Amt bits

Each bit enables a power-of-two stage (1, 2, 4...).

Modes

Logical / arithmetic shift vs rotate (wrap).

Barrel shifter starter

Workbook practice

- In the workbook track, SLL hex eight zero right by one and give the result
- For amount five, list which power-of-two stages fire
- Sketch three mux layers for an eight-bit barrel
- Contrast SRL and SRA on a negative pattern
- Name one pitfall: confusing barrel stages with counting single-bit shifts one at a time

Pitfalls to watch

- Do not assume amount N means N serial one-bit layers
- SRA and SRL differ only in the fill bit
- And remember: the browser lab is literacy
- Real ALUs still need variable-width rules, flags

Your turn

- Complete the checklist for at least one track, preferably both
- In the browser, finish a few challenges after the starter
- On paper, trace one starter stage by stage
- When you are ready, take the short quiz, then continue to seven segment